EmbyConvert on UGREEN DXP NAS (Docker)

Manual (EN) – Version v1.1 – As of 2026-01-22

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1. Introduction

This package provides a HEVC (H.265) batch encoder for media libraries. It is designed to automatically scan an existing Emby input folder (e.g., Movies/Series) and write compatible files to a separate output folder as MKV with HEVC video.

The encoder uses Intel Quick Sync (QSV) via /dev/dri. This makes the package suitable for UGREEN DXP models (Intel iGPU available). DH series models are not supported.

2. Features

- Automatic scan of the input directory (multiple video formats: MKV/MP4/M4V/WEBM/AVI/MOV/TS/M2TS).
- Converts non-HEVC videos to HEVC (H.265) and writes them as .mkv to the output (folder structure is preserved).
- Safety mode: optional input scan for hard errors (STRICT_INPUT) and output verification (VERIFY_OUTPUT).
- Size guard: by default, the output must not be larger than the source (with optional retry logic if needed).
- Conservative by default: 4K, HDR and Dolby Vision are skipped (can be changed via ENV).
- Resume/State: manifest + progress are stored in the output directory so restarts can continue cleanly.
- Watch mode: repeats the scan at fixed intervals and processes new files automatically.
- Optional: remux already-HEVC files to MKV as well as generate a HEVC candidate list (and optional re-encode).
- Logs: central run logs in the output (e.g., /out/_ffmpeg_logs) and skip lists for troubleshooting.
- Additional report: script to find MP4/MKV duplicates (nothing is deleted automatically).
- Optional: weekly email report (SMTP) on Friday at 20:00 with a list of converted files and space savings (Input/Output/Δ/%).

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3. Package contents and folder structure

Included in the package:

- docker-compose.yaml – Starts the FFmpeg container incl. QSV and runs the script in watch mode.
- ffmpeg/scripts/hevc_batch.sh – Main script (batch encoder, resume, watch mode, safety checks).
- ffmpeg/scripts/report_mp4_mkv_dupes.sh – Duplicate report (MP4 + MKV with the same base name).
- ffmpeg/scripts/weekly_mail_report.py – Generates the weekly report (HTML/TXT) and optionally sends it via SMTP.
- ffmpeg/scripts/weekly_mail_report_daemon.py – Small scheduler that runs the report automatically on Friday at 20:00.
- embyconvert_mail.env – Template for SMTP/report settings (located in the same folder as docker-compose.yaml).

The recommended location is a folder on a data volume, e.g. under:

```
/volume2/docker/EmbyConvert/
```

4. Requirements

- UGREEN DXP NAS (Intel iGPU / Quick Sync available).
- UGOS Pro with Docker/Container Manager and Docker Compose support.
- An existing input folder with media (e.g. share "Emby").
- A separate output folder with enough free space (e.g. share "Embykonvertiert").
- The Docker user (PUID/PGID) needs write permissions in the output folder.

5. Preparation: IN/OUT shares and paths

5.1 IN path (example: "Emby")

The IN path is your existing media share. In your example it is called "Emby" and contains movies and series. This path is mounted into the container as /in (read-only recommended).

Example (UGREEN default paths – please check which volume you are using):

```
IN  (Host): /volume1/Emby
IN  (Container): /in (read-only)
```

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5.2 Create OUT path (example: "Embykonvertiert")

Create a new shared folder in UGOS (e.g. "Embykonvertiert"). This folder is the target path for converted files and is mounted into the container as /out.

Important:

- The output must be writable (PUID/PGID must have permissions).
- The script automatically creates subfolders for logs/state in the output: _ffmpeg_logs and _state.
- The original files in the input are not modified or deleted.

Example paths:

```
OUT (Host): /volume1/Embykonvertiert
OUT (Container): /out
```

5.3 What ends up in the OUT folder?

- Converted files: same relative path as in the input, but as .mkv.
- Logs: /out/_ffmpeg_logs/ (e.g. run_ALL.log).
- State/Resume: /out/_state/ (manifest, progress, skip lists, optional reports).

6. Setup

6.1 Copy the files

Example (SSH/terminal on the NAS – adjust paths):

```
mkdir -p /volume2/docker
cd /volume2/docker
unzip EmbyConvert.zip -d /volume2/docker
cd /volume2/docker/EmbyConvert
```

Afterwards, docker-compose.yaml should be located directly in /volume2/docker/EmbyConvert and the scripts under ./ffmpeg/scripts/.

6.2 Adjust docker-compose.yaml

Open docker-compose.yaml and adjust at least the volume mounts for /in, /out and /scripts (according to your shares and your storage location).

Important lines (example):

```
volumes:
  - /volume1/Emby:/in:ro
  - /volume1/Embykonvertiert:/out
  - /volume2/docker/EmbyConvert/ffmpeg/scripts:/scripts
```

Optional/Recommended:

- Set PUID/PGID correctly (Docker user that can write to /volume2/docker and the output).
- Keep devices: /dev/dri:/dev/dri enabled (QSV).
- group_add for video/render may be required depending on NAS firmware/permissions.

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6.3 Start

In the folder /volume2/docker/EmbyConvert:

```
docker compose up -d
docker compose ps
```

View live log:

```
docker compose logs -f ffmpeg
```

Note: On the first run, the initial scan may take a while depending on the library size. In watch mode, the script will then regularly look for new files.

6.4 Optional: Weekly email report (Friday 20:00)

Optionally, the package can send a report via SMTP once per week. The report lists only files that were actually converted (OUT/*.mkv with a matching source in IN) and shows space savings in an Excel-like style.

Step 1: Configure embyconvert_mail.env

The file embyconvert_mail.env is located in the same folder as docker-compose.yaml. Enter SMTP_HOST, SMTP_PORT, SMTP_USER, SMTP_PASS as well as SMTP_FROM/SMTP_TO. REPORT_TZ should be set to Europe/Berlin.

Step 2: Start the weekly report service

In the folder /volume2/docker/EmbyConvert:

```
docker compose up -d weekly_mail_report
docker compose logs -f weekly_mail_report
```

Test (send email immediately):

```
docker exec -it weekly_mail_report python3 /scripts/weekly_mail_report.py
--in /in --out /out --envfile /config/embyconvert_mail.env --force-send
```

You can find report files and logs in the output under /out/_ffmpeg_logs/ (HTML/TXT + weekly_mail_report.log).

7. How it works (brief)

7.1 Scan, manifest and resume

On startup, a manifest with all found video files is created and stored in the output under /out/_state/manifest_ROOT.txt. Progress is stored in /out/_state/progress_ROOT.txt. After container restarts, processing automatically continues from the last position (resume).

Rebuild the manifest (e.g. if many files were moved/renamed):

```
docker exec -it ffmpeg_qsv /scripts/hevc_batch.sh --rebuild-manifest
```

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7.2 Standard conversion (non-HEVC -> HEVC)

- By default, the script encodes only the first video stream (v:0) to HEVC (hevc_qsv).
- Audio is copied by default (no re-encode), as are chapters/metadata; the container format is MKV.
- Hybrid mode: first ICQ (quality), then optionally bitrate mode as a fallback.

7.3 HEVC input (already H.265)

- Default: already-HEVC files are skipped (REMUX_HEVC=0).
- Optional: REMUX_HEVC=1 remuxes HEVC files to MKV (copy) if no output exists yet.
- Optional: the script can mark HEVC files as candidates for re-encode (CSV in /out/_state/).

7.4 Safety features

- STRICT_INPUT=1: ffmpeg scans the source for typical corruption patterns; files with errors are skipped.
- VERIFY_OUTPUT=1: the output is scanned again after encoding; faulty outputs are discarded.
- SIZE_GUARD=1: the output must not be larger than the input; if it is, depending on the settings it is skipped or retried.
- Skip lists: problematic files are written to /out/_state/skip_*.txt and will be skipped automatically on the next run.

8. Important settings (ENV in docker-compose.yaml)

Most important parameters (excerpt):

Variable	Meaning (brief)
GQ / PRESET	Quality/speed for QSV (ICQ and preset).
SKIP_4K / SKIP_HDR / SKIP_DOLBY_VISION	Protection: skip 4K/HDR/DV by default.
SIZE_GUARD / ALLOW_SIZE_GROW_PCT	Prevents larger outputs (default: 0% growth).
STRICT_INPUT / VERIFY_OUTPUT	Input scan + output verification for more safety.
EXCLUDE_DIRS	Ignore folders (e.g. #recycle, @eaDir).
WAIT_SEC	Interval in watch mode (seconds).
REMUX_HEVC / REENCODE_HEVC	HEVC handling: remux and candidates/re-encode.
AUDIO_GERMAN_ONLY	Optional: if multiple audio tracks exist, keep German only (otherwise skip if unclear).

Tip: If you also want to convert 4K/HDR, change the SKIP_* variables intentionally and test with a few files first.

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9. Report: Find MP4/MKV duplicates

After a while you may have both originals (.mp4) in the input and converted files (.mkv) in the output. The script report_mp4_mkv_dupes.sh helps to find such duplicates. It does not delete or move anything.

Example call (creates CSVs in /out/_state):

```
docker exec -it ffmpeg_qsv /scripts/report_mp4_mkv_dupes.sh /out /out/_state
```

10. Weekly report via email (SMTP)

The weekly report is optional and runs by default every Friday at 20:00 (Europe/Berlin). It only considers files that were newly created/updated as .mkv in the OUT directory and for which a matching source exists in the IN directory (same relative path + same file name, different extension).

Calculation (like Excel):

- Input = size of the source file in IN
- Output = size of the converted file in OUT
- Δ (delta) = Output - Input (negative means space saved)
- % = Δ / Input (negative means percentage space saved)

Important ENV variables (embyconvert_mail.env):

- SMTP_HOST, SMTP_PORT, SMTP_TLS, SMTP_SSL
- SMTP_USER, SMTP_PASS (optional, if authentication is required)
- SMTP_FROM, SMTP_TO (multiple recipients via comma or semicolon)
- REPORT_DAYS (default 7), REPORT_MAX_ITEMS, REPORT_SUBJECT_PREFIX
- REPORT_DRY_RUN=1 writes HTML/TXT only, but does not send an email

Note: To avoid spam, the report is sent at most once every ~20 hours by default. For testing you can use --force-send.

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11. Troubleshooting

- No QSV / no hardware acceleration: Check whether /dev/dri exists on the NAS and is mounted into the container.
- Permission issues: Check PUID/PGID and write permissions on the output; adjust group_add for video/render if needed.
- Note: 'groups: cannot find name for group ID 105' in the container is usually harmless (just a missing group name).
- Nothing happens: Check /out/_ffmpeg_logs/run_ALL.log and the manifest at /out/_state/manifest_ROOT.txt.
- Many files are skipped: the cause may be SKIP_4K/HDR/DV or STRICT_INPUT (see skip lists).
- Output is discarded: VERIFY_OUTPUT reports errors or SIZE_GUARD triggers (details in the log).

12. Safety notes

- Mount the input as read-only (:/in:ro) so originals are never modified.
- Make the output share writable only for the required users/apps.
- Conversion creates high load (CPU/iGPU, storage). Use time windows or throttle via preset/parameters if needed.

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